Blood collection tubes contain a vacuum that helps draw the proper amount of blood. Each tube that contains an additive has a unique stopper color that identifies the additive placed in the tube. These additives can be in the form of liquid or powder and also have an impact on what type of sample they create when collected from the body and centrifuged. Typically red, gold, or serum separator tubes will yield serum, while most other tubes yield plasma. Serum tubes need to clot for 30 to 60 minutes before being centrifuged. Plasma may be centrifuged immediately after collection. After being centrifuged, both serum and plasma samples will look the same, so it is critical to use the proper tube for every venipuncture.

Tubes are collected in a particular order called the order of draw. Attached to the double-ended needle, the side that enters the tube has a cover that slides up when the tube is attached. When the next tube is attached, some of the additives can be carried from the tube previously collected to the next tube. The laboratory has their equipment calibrated to process tests in a particular order. When a tube is collected in the improper order, the additional additives may change not only how the blood is preserved but also the blood test results. This can negatively impact your patient's diagnosis and treatment. It is important to always follow the order of draw specified by your facility's laboratory.

TABLE 4.3 The Order of Draw of Commonly Used Tubes

VACUUM TUBE COLOR	ADDITIVE	LABORATORY USE	NUMBER OF INVERSIONS
Yellow-top tube or blood culture bottles	• Sodium polyanethol sulfonate • Prevents blood from clotting and stabilizes bacterial growth	Blood or body fluid cultures	-
Light blue	• Sodium citrate • Removes calcium to prevent blood from clotting	Coagulation testing	3 to 4
Red	None	• Serum test • Chemistry studies • Blood bank • Immunology	5

TABLE 4.3 The Order of Draw of Commonly Used Tubes

VACUUM TUBE COLOR	ADDITIVE	LABORATORY USE	NUMBER OF INVERSIONS
Red/gray marbled	• No anticoagulant but contains silica particles to enhance clot formation • For serum separation	• Serum test • Chemistry studies • Immunology	5
Green	• Heparin • Inhibits thrombin formation to prevent clotting	Chemistry test	8
Green/gray marble	• Lithium heparin and gel • For plasma separation	Plasma determinations in chemistry studies	8
Lavender	• Ethylenediaminetetraacetic acid (EDTA) • Removes calcium to prevent blood from clotting	Hematology test	8
Gray	• Potassium oxalate and sodium fluoride • Removes calcium to prevent blood from clotting • Fluoride inhibits glycolysis	Chemistry testing, especially glucose and alcohol levels	8 to 10

**Always follow manufacturer recommendations when determining the order of draw.*

When choosing the appropriate tubes for your blood collection, also check them for their expiration date. The vacuum is reduced in expired tubes, which could result in a lower volume of blood collected. Another issue with expired tubes is that the additive may be expired. Expired additives may adversely affect the test results. Both issues can contribute to a patient having to endure more testing or receiving an inaccurate diagnosis and treatment because of test results that are not in range.

Do Not Use Expired Vacuum Tubes

When a tube expires, the amount of vacuum and quality of the additives are reduced.